



Collaboration with Highly Automated Vehicles via Voice Interaction and Augmented Reality: A VR-Based Study

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ABSTRACT

In future confined industrial contexts (hubs), highly automated vehicles and human operators may work in shared spaces and collaborate on joint tasks. This will probably generate a demand for new user interfaces between humans and machines that need to be designed to facilitate high levels of safety and efficiency as well as a positive user experience (UX). The present work investigates the potential of using a combination of voice interaction (VI) and visual augmented reality (AR) to support collaboration between automated vehicles and humans manually operating a machine. A concept using VI and AR for a loading scenario in a logistic center was created and evaluated using a VR headset to provide an immersive experience. A user study with 18 forklift drivers was conducted. Our study shows that the concept generated high scores in terms of usability and UX, which indicates a promising potential to use VI and AR to facilitate interaction between human machine operators and unmanned highly automated vehicles when performing collaborative tasks. Our study also implies a need to explore the design and implementation of more complex and social VI for users in logistic centers.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)** → **Interaction techniques**

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KEYWORDS

voice interaction, augmented reality, self-driving vehicles, autonomous vehicles, natural user interface, interaction design

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1 BACKGROUND

The possibility of using highly automated heavy vehicles in confined industrial contexts (hubs) has gained a lot of interest in industry in recent years. This technical development may contribute to more efficient production, safer work areas, and a reduced environmental footprint in contexts such as logistic centers and underground mines [1, 2]. In future hubs, highly automated vehicles (SAE levels 4 and 5 [3]) and human operators may work in shared spaces and collaborate on joint tasks. Considering that vehicle drivers are no longer present, new types of user interfaces between humans and machines need to be designed to facilitate high levels of safety and efficiency as well as a positive user experience (UX). The present work investigates the potential of using voice interaction (VI) and visual augmented reality (AR) to support collaboration between automated vehicles and humans manually operating a machine. A pre-study in the research project In the Hub [4] identified loading tasks as situations that currently require complex interactions between machine operators and vehicle drivers. Thus, supporting collaboration between forklift operators and unmanned trucks was selected as a particularly interesting use case to study and thereby learn more about the appropriateness of the interaction technologies.

Most information in modern work machines is presented on displays and tablets, and interaction is typically performed using physical controls and elements provided on screens (e.g., virtual buttons). Exploring emerging interaction technologies, such as VI, in hubs could generate new opportunities to improve UX for more groups of people (e.g., professional workers in hubs), making new technologies more widely accessible.

Also relying on specific interaction technologies that fit most users involves the risk of limiting accessibility for other users. Thus, examining the usefulness of alternative interaction technologies will increase our understanding of how these solutions can meet the needs of users with limited abilities.

VI is the natural way to communicate between humans and offers the possibility of interacting while keeping hands and eyes free. Previous research has shown that using VI can facilitate interactions between humans and automated vehicles [4, 5, 6, 7]; however, most work has focused on applications inside cars and interactions with pedestrians.

While visual information is typically presented on conventional screens and tablets, recent work has examined the use of AR to support human operators in various industrial environments [8]. Researchers have suggested that the technology can facilitate operators' productivity, safety, and confidence [9]. By using AR and place information near the operators' line of sight, workload may be reduced [10]. AR also offers the possibility of virtually placing information directly in the work scene (e.g., on a vehicle or pallet), which can facilitate the workers interpretation and use of the information.

Combining VI and AR in interaction models has been explored to some extent—for instance, for loader cranes [11]. However, we still know little about how these technologies can be used and combined to facilitate collaboration with highly automated vehicles in work tasks. A recent study by Fagerlönn et al. [2022] examined the combination for use in interactions with highly automated vehicles in mines and logistic centers. The study showed high potential in both technologies, as well as in their combination, to facilitate high acceptance and work efficiency. However, the study used a video-based methodology, and the researchers highlighted the importance of evaluating the technologies in more realistic interactive test setups.

1.1 Aim

The aim of the present study was to investigate the potential of using VI and AR in combination to facilitate usability and UX in collaborations between professional human operators and unmanned highly automated vehicles in logistic center contexts by using an interactive prototype and test setup.

2. METHOD

2.1 Concept

Overall, the information content of the concept was based on the work presented by Fagerlönn et al. [2022]. Figure 1 shows the AR screen of the concept along with explanations of the different AR elements.

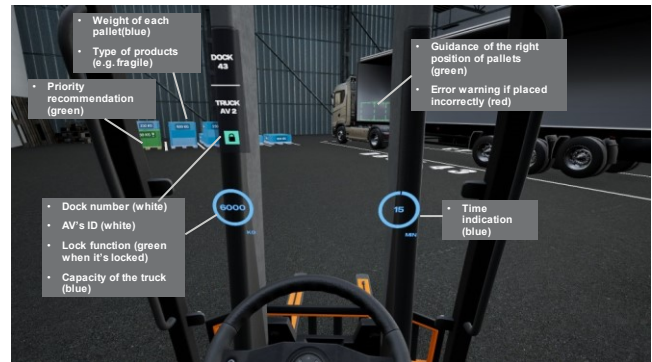


Figure 1: AR elements in the concept. Grey boxes provide explanations to the different elements.

The screen is projected on the windscreen of the forklift in the VR environment and contains both 2D and 3D elements. To reduce the risk of AR elements obscuring objects in the work scene, 2D elements are placed in front of the masts of the forklift. The 2D elements show remaining load capacity in the truck, remaining time to departure, the name of the loading dock, and the name of the truck. Also, an indication of the vehicle listening to the operator's voice was presented. The 3D elements indicate which pallet to load (marked in green), information about the pallet (e.g., weight) and where to place it in the compartment. Misplaced pallets in the compartment are also highlighted.

The VI consisted of information and questions provided by the truck. To make it as easy as possible to interact with the truck, the language of the operator's area of operation was used (Swedish). A female voice was used to represent the truck. Initially, the truck greets the operator and provides information about the position of the vehicle in the logistic center, total load capacity and time to departure. The trucks ask questions to prepare the loading task (e.g., if it should open the load compartment). When the load capacity is reached, the truck informs the operator and asks if it should close the compartment and unlock itself. The operator can reply to the questions using either simple answers (e.g., "yes") or a bit more complex answers (e.g., "please do that").

The concepts and virtual environment were created using the Unreal Engine software (Epic Games Inc, USA). VI was integrated using Google API [13], as it is one of the most reliable APIs and it facilitates the development for this particular case.

2.2 Participants and apparatus

Eighteen professional forklift operators participated in the study. Their mean age was 37.6 years ($SD = 9.2$). The participants experienced the concept using a Varjo Aero VR headset (Varjo Technologies Oy, Finland) and headphones. The forklift was controlled by the subjects using a Logitech G920 wheel and pedals setup (Logitech International S.A, Switzerland). Operating the forklift was a simplification of driving a real forklift, but it was possible to move forwards, backwards, break and lower/higher the

forks. To communicate with the truck using voice, the subject pushed a button on the steering wheel.

2.3 Procedure and measurements

The subjects sat in a chair which was adjusted to the participants height so they could comfortably use the steering wheel pedals when driving the forklift in the VR environment. A script was followed during the sessions, so all participants got the same information, and everything presented in the same order. A brief warmup session was performed, without the VI and AR elements, before the real test to allow the subject to familiarize with the controls and VR environment.

The loading scenario was around 8 min long and subjects typically performed the scenario in VR two times. Some subjects did the scenario in VR only once due to motion sickness. To reduce the risk of simulator sickness, there was a break between in the scenarios. The scenario was performed in a virtual representation of a logistic center. The truck to be loaded was parked near several pallets. The scenario contained the steps (1) to prepare for loading, (2) load the truck with 5 pallets, and (3) prepare the truck for departure. In the first step, the operator and vehicle performed a conversation to lock the truck (so that only the operator could control the truck) and open the load compartment. The subject also received information about e.g., load capacity and departure time. In the second step, the operator was guided in AR on which pallet to load and where to place it in the load compartment. In the third step, a conversation was held to close the compartment and unlock the truck to allow departure.

After performing the scenario once, the subjects were asked to rate their experience using the System Usability Scale (SUS) [14]. After performing the scenario two times (or one time in case they experienced too much simulator sickness), they rated their experience using the User Experience Questionnaire (UEQ) [15]. Acceptance of each interaction technology was assessed using the method by Van der Laan et al [1997]. The subjects also answered statements about their experience using scales ranging from 1 (do not agree at all) to 7 (completely agree). In the end, a semi structured interview was conducted to get a deeper understanding of the UX. In addition to UX assessments, simulator sickness was assessed after the scenarios using the Motion Sickness Assessment Questionnaire (MSAQ) [17].

3 RESULTS

3.1 Overall system usability and user experience

The SUS scale ranges from 0 to 100. The concept generated a mean SUS score of 75.0 (SD = 17.4). According to Bangor et al. [18], the result corresponds to the adjective “good” on a scale from “worst imaginable” to “best imaginable”.

Table 1 shows results of the UEQ questionnaire. While the SUS scale focus especially on usability, the UEQ is designed to assess a broader aspect of UX. The questionnaire consists of 26 items, which are used to determine the attractiveness, pragmatic quality (perspicuity, efficiency, dependability) and hedonic quality (stimulation, originality) of the system. The UEQ scores range from

-3 (horribly bad) to 3 (extremely good). Scores between -0.8 and 0.8 is considered a neutral evaluation and values above 0.8 represent a positive evaluation.

Table 1: UEQ scores and 95 % confidence intervals (CI).

	Mean score	CI
Perspicuity	1.5	1.3
Efficiency	1.2	1.7
Dependability	0.9	1.1
Stimulation	1.5	1.4
Novelty	1.1	1.4
Attractiveness	1.4	0.9

Overall, the concept generated scores corresponding to positive UEQ evaluations. It should be noted that according to [15], it is extremely unlikely to observe mean values above 2.0. The highest scores were observed for the scales Perspicuity and Stimulation. Perspicuity includes properties such as being easy to understand, simple, and easy to learn. The scale Stimulation includes properties such as being interesting, motivating, and exciting. The lowest score (still positive) was observed for the scale Dependability, which includes words such as predictable, supportive, and secure. Here, it should be mentioned that in the interviews, there were a few comments about trusting a machine or computer, that one can feel insecure. Also, there were comments about how the system would work if something unexpected (not planned) would happen, or if you do not respond as intended to the system.

In the interviews, the operator’s comments were generally positive about the concept. Unmanned heavy vehicles, VI and AR are all new technologies, but they thought that you would learn and get into it eventually, even if it will depend on the person how adaptable you are. However, they also believed that the solution belongs to the future rather than the present, as some development is still required to make it work well. A contributing reason for this may have been that eight subjects (44%) had difficulties to communicate without any interference using VI, e.g., they needed to repeat a message or rephrase it at least once. This indicates that it is of great importance to develop VI solutions that can accurately interpret the voice messages, and support users to recognize, diagnose, and recover from errors.

Moreover, it was mentioned by participants that they often work towards a deadline and misunderstandings, or incorrect information, takes valuable time from the loading. It is important that the communication is time effective to improve efficiency at work. To work with the concept was considered to save time because of the instant information, but at the same time considered by some to make the work less stressful since the phase could be slower. This indicates a need to design a proper flow for the collaboration. It is probably good to have a well-balanced communication flow and phase, so that users do not need to rush to meet the deadlines or slow down without any reason.

3.3 Acceptance of interaction technologies

The methodology by Van der Laan et al. [1997] consists of a set of rating items which are used to access the usefulness and satisfaction of the system. The scales range from -2 to +2. The VI resulted in a high mean usefulness score of 1.3 (SD = 0.7) and satisfaction score of 1.1 (SD = 0.7). The statement “I can imagine working with self-driving trucks that communicate by voice on a daily basis in the future” received a high mean score of 5.7 (SD = 1.1), and the statement “Communicating via voice is a good way to cooperate with the truck” received a mean score of 5.8 (SD = 1.7).

Compared to the VI, the AR resulted in somewhat higher mean acceptance scores. The AR solution received a usefulness score of 1.6 (SD = 0.4) and satisfaction score of 1.5 (SD = 0.6). The statement “I can imagine working with self-driving trucks that communicate by AR on a daily basis in the future” received a mean score of 6.2 (SD = 1.2), and the statement “Communicating via AR is a good way to cooperate with the truck” received a mean score of 6.3 (SD = 1.0).

In the interviews, there were various opinions about talking to a computer. Some expressed it could become boring, and some that it would be less misunderstandings due to the fact that there is no human driver involved in the communication.

4 DISCUSSION

Overall, the concept generated high scores in terms of usability and UX. The results support the notion that VI and AR have promising potential to facilitate interaction between human machine operators and unmanned highly automated vehicles when performing collaborative tasks. These results are in line with previous research conducted by Fagerlön et al. [2022]. However, unlike that study, the present study examined users’ experiences in an interactive and immersive VR setup, that was designed to produce a more realistic experience. In addition to the study’s results, the proposed interaction model should be valuable input for researchers and practitioners who aim to examine and develop solutions for collaborative scenarios.

Both AR and VI generated high acceptance scores. The subjects seemed especially positive about the use of AR, whereas VI tended to receive somewhat lower scores. These results indicate that AR could be especially useful in future implementations. However, the intention of the study was to evaluate the combination of the technologies, not compare the technologies.

It should also be noted that the technical possibilities of implementing dynamic VI in the interactive environment were limited in the present work. The interaction consisted of information and questions provided by the vehicle; the user only needed to reply using simple answers. This type of question-based interaction may be suitable and helpful in making users recognize rather than recall, which is considered an important design guideline for a voice user interface [19]. However, it has been suggested that more conversational VI could lead to higher levels of pleasantness and trust [6]. Fagerlön et al. [2022] reported that the preference for more social dialogue seemed to vary among different types of hub contexts. For instance, many forklift operators seemed to appreciate a more social human-machine dialogue over simple

voice commands. Our results also reflect differences in individual preferences. Some individuals appreciated the short conversations, whereas others thought it would be good to be able to ask questions or give longer commands. Thus, future research should aim to implement and investigate VI that allows more complex and social dialogues than was possible in the present concept, to provide more alternatives for the users.

The results should be interpreted with caution. First, some subjects had trouble making the system understand what they said; they had to repeat their answers, and this difficulty might have impacted their evaluations. Also, the study had a rather small sample size and only investigated the solution in one pre-defined scenario. In addition, experiencing solutions in a VR context is not the same as experiencing them in reality. In the future, we hope to implement and examine solutions in a real hub. Finally, the study’s end users were all working in Sweden, and the cultural context might have impacted the results.

5 CONCLUSION

Our study investigated the potential in using a combination of VI and AR to support collaboration between automated vehicles and humans manually operating a machine. A concept combining VI and AR for a loading scenario in a logistic center was designed and evaluated with forklift operators. The results suggest that VI and AR combined have high potential for facilitating the interaction between humans and unmanned machines. AR scored somewhat higher in terms of acceptance than VI, indicating that AR could be especially useful for future implementations. However, the aim of the study was not to compare the relative appropriateness of VI and AR. Additionally, both previous research [12] and our study implies a requirement for further researching and exploring more social voice interfaces for users in logistic centers.

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